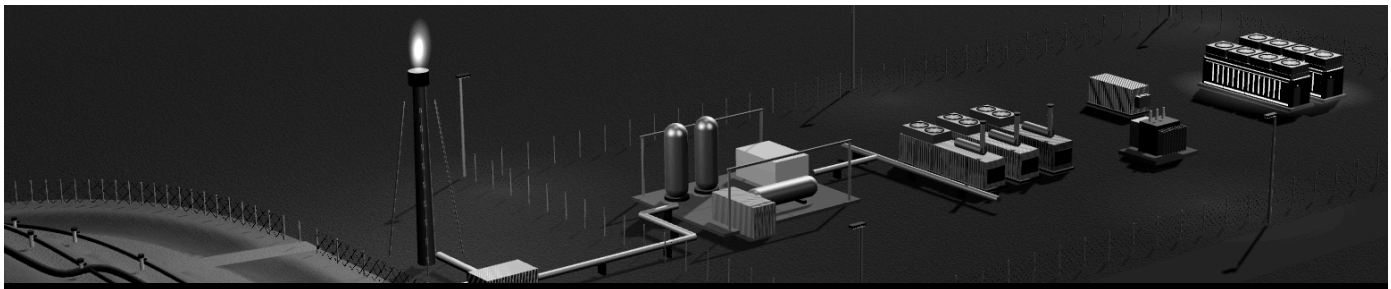
Waterloo Landfill Site (Erb Street West), Waterloo, Ontario: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

2.2 MW	2	8 ac	204k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $153k	$8k to $15k	$8k to $168k



YOUR SITE IN OUR LIVE 3D MODEL ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=877&CH4=46.4

THE MATH

Landfill gas collected, used by the engine plant until May 2027 (2025)	877 scfm
Methane content	46.4%
Energy in the gas	24.5 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 2.2 MW
Compute (power less cooling and losses, /1.2)	about 1.9 MW
Compute modules (about 1 MW each)	2
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 8 acres
Gas bought per year (95% run time)	204k MMBtu
Gas payment, up to $0.75/MMBtu	up to $153k
Land lease at $1,000 to $2,000 per acre per year	$8k to $15k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Toromont's IESO contract ends 2027-05-17: whether its engines, switchgear and grid tie stay and can be reused
- MECP Environmental Compliance Approval (air and noise) under EPA s. 9; O. Reg. 359/09 REA exemption for host-site plants of 10 MW or less
- Part V approval amendment for the gas system; O. Reg. 232/98 landfill gas duties
- City of Waterloo zoning and site plan control; Region lobbyist registry (By-law 24-043) within 10 business days of first contact

Source: Ontario landfill gas dataset, 2025: about 877 scfm collected at 46.4% methane, almost all used by the 6.2 MW engine plant. IESO's contract list shows that plant's contract ending May 17, 2027. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.